Financial Institutions & Markets- ENT 309

PPT #5, IIITD, Summer, 2026

Agenda

- Why Bonds have different interest rates?
- Yield Curve
- Why do Interest Rates change?
- Supply, demand and equilibrium for bonds
- Movement along the curve and Shift of the curve

Real & Nominal Interest

- Inflation Rate: Rate of change in prices. %
- Nominal Interest Rate = Real Interest Rate **PLUS** Impact of <u>expected</u> Inflation

$$i = i_r + \pi^e$$

- All are in Percentages

$$1 + i = (1 + i_r)(1 + \pi^e)$$

- Since the last term $(i_r \times \pi^e)$ is small, it is ignored.
- In most instances, interest rate would mean nominal interest rate. Unless specified otherwise.

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Why do Bonds have different Interest Rates?

Why Diff. Interest Rates?

Name of the Issuer	Maturity	Yield
Spandana Sphoorty	Apr 2028	12.44%
Finnable Credit	Aug 2028	11.75%
Namra Finance	Sep 2028	11.66%
Berar Finance	Sep 2028	11.25%
Early Salary	Jul 2028	11.20%

- https://www.indiabonds.com/search/
- Why do bonds with same maturity have different yields?
- To take into account (a) Default Risk (b) Liquidity Risk. Refer Bond Rating discussion earlier.

Why Diff. Interest Rates?

Issuer: Government of India				
Maturity	Yield		Maturity	Yield
Sep 2030	6.6%		Jul 2037	6.96%
Oct 2034	6.79%		Aug 2054	7.56%
May 2035	6.85%		Feb 2061	7.58%
May 2036	7%		Apr 2065	7.59%

- https://www.indiabonds.com/search/
- Why do bonds with different maturity have different interest rates?

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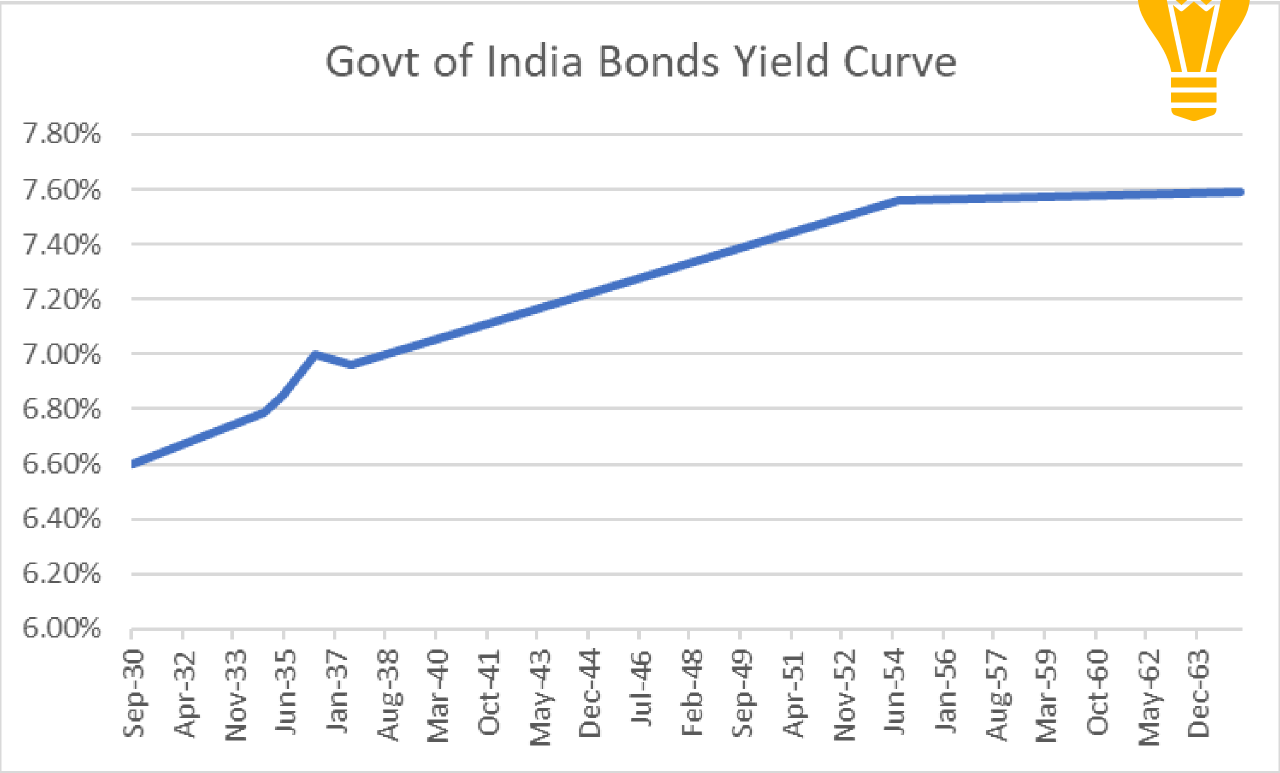
Yield Curve

Drawing Yield Curve



- Drawn using data on earlier slide.
- Why is the shape of the Yield Curve like this?

Drawing Yield Curve

- Liquidity Preference Theory: TWO components determine long-term int rate
  - (1) The interest rate on a long-term bond will equal an average of short-term interest rates expected to occur over the life of the long-term bond
  - (2) PLUS a 'liquidity premium'.

$$i_{nt} = \frac{i_t + i^e_{t+1} + i^e_{t+2} + \cdots + i^e_{t+(n-1)}}{n} + l_{nt}$$

Where

$$\begin{aligned} i_t &= \text{year 1 interest rate} \\ i^e_{t+1} &= \text{year 2 interest rate} \\ l_{nt} &= \text{liquidity premium} \\ n &= \text{number of years} \end{aligned}$$

Drawing Yield Curve

Liquidity Premium

- Investors prefer shorter-maturity bonds- they carry less int rate risk.
- To make them buy longer term bonds, a ‘liquidity premium’ is required.
- Liquidity: How easily can a Bond holder sell/convert to cash. Short term bonds are more liquid: Can be sold more easily as compared to long term bonds because short term bonds carry less int rate risk.
- Liquidity premium is **positive and increases** with maturity.
- Usually, Yield curve is upward sloping. It is expected that short term interest rates will go up.
- In other situations, the Yield curve can be downward sloping or flat.

Yield Curve

Let us suppose that the one-year interest rates over the next five years are expected to be 5%, 6%, 7%, 8% & 9%. Investors' preference for holding short-term bonds have the liquidity premiums for oner-year to five-year bonds as 0%, 0.25%, 0.5%, 0.75% and 1% respectively. What is the interest rate on a five-year bond? For a two-year bond?

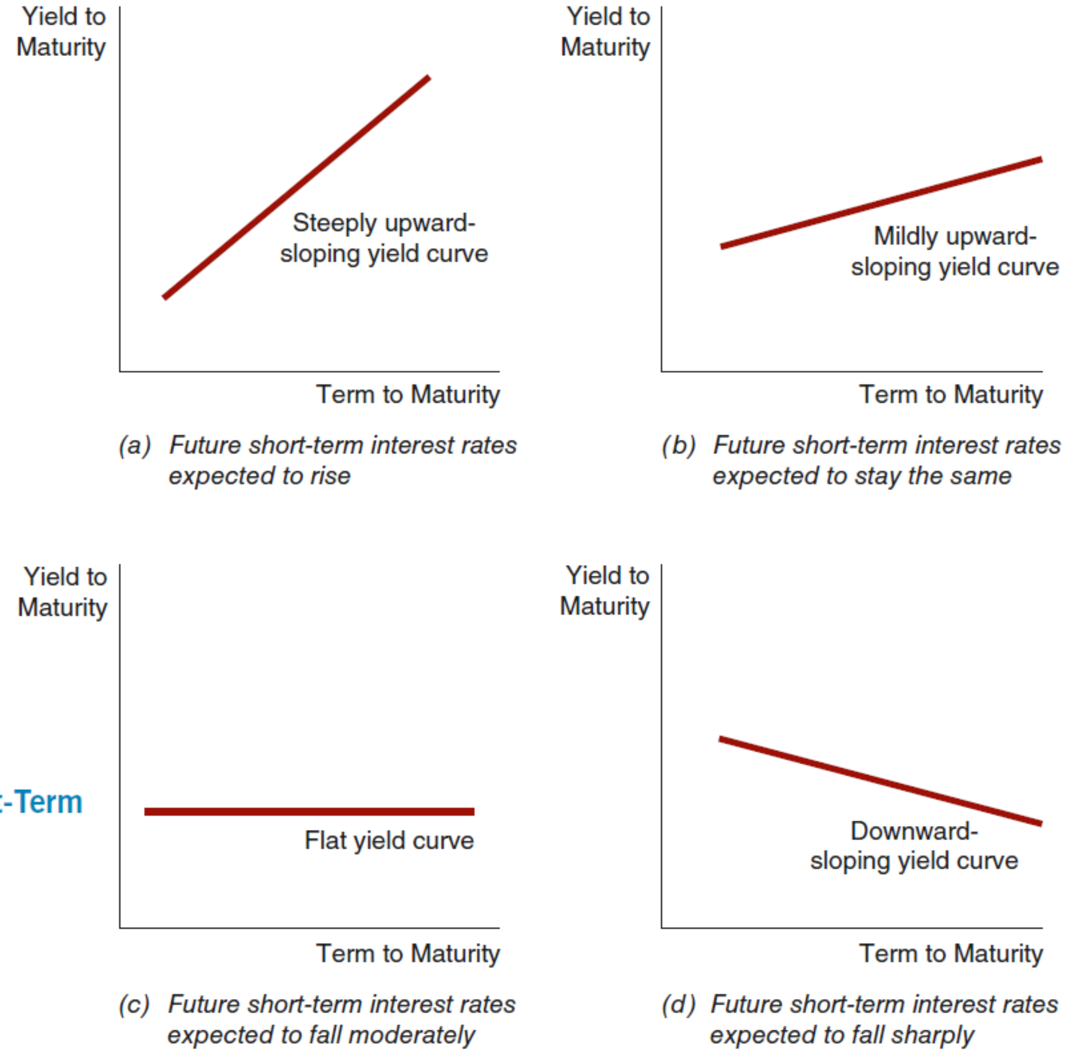
$$i_{5t} = \frac{5\% + 6\% + 7\% + 8\% + 9\%}{5} + 1\% = 8.0\%$$

$$i_{2t} = \frac{5\% + 6\%}{2} + 0.25\% = 5.75\%$$

Yield Curve

- When will the yield curve be downward sloping?
    - Downward sloping yield curve is called INVERTED yield curve
    - When future interest rates are expected to go down sharply.
    - & This effect overcomes the positive ‘liquidity premium’ effect
- When will it be flat?
    - When future interest rates are expected to go down slightly.
    - & This effect balances the positive ‘liquidity premium’ effect
- Even if future interest rates are expected to remain flat (unchanged), the yield curve will be upward sloping due to the positive ‘liquidity premium’ effect,

Yield Curve



Yield Curves and the Market's Expectations of Future Short-Term Interest Rates According to the Liquidity Premium Theory

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Demand & Supply of Bonds & Equilibrium

Change in Int Rates

- Why do interest rates change?
- Why should we be interested in predicting change in Int Rate?
- Bond prices are inversely related to interest rates. So, we can study change in bond prices to understand change in interest rates.
- We study the demand and supply of bonds. And derive the equilibrium prices and quantity.

Demand for an asset/ Bonds

- Asset is something which has value and can give returns

4 main factors determine the demand for a particular asset

- Wealth of an individual
- Expected return of one asset as compared to another asset
- Risk of one asset relative to another.
- Liquidity (ease of selling to convert into cash) relative to another asset
- There could be other factors also. We discuss 4 main ones.

Demand for an asset/ Bonds

- Expected return: Expected return on an asset is the weighted average of all possible returns, where the weights are the probabilities of occurrence of that return

$$R^e = p_1 R_1 + p_2 R_2 + \cdots + p_n R_n$$

where

$$\begin{aligned} R^e &= \text{expected return} \\ n &= \text{number of possible outcomes (states of nature)} \\ R_i &= \text{return in the } i\text{th state of nature} \\ p_i &= \text{probability of occurrence of the return } R_i \end{aligned}$$

- Risk is the uncertainty of an asset's returns.
  - Measured by the Standard Deviation of the returns,
  - The higher the standard deviation, the greater the risk of an asset.

Demand for an asset/ Bonds

- Wealth: Holding everything else constant, an increase in wealth raises the quantity demanded of an asset. POSITIVELY RELATED.
- Expected Return: an increase in an asset's expected return relative to that of an alternative asset, holding everything else unchanged, raises the quantity demanded of the asset. POSITIVELY RELATED.
- Risk: Holding everything else constant, if an asset's risk rises relative to that of alternative, its quantity demanded will fall. NEGATIVELY RELATED.
- Liquidity: The more liquid an asset is relative to alternative assets, holding everything else unchanged, the more desirable it is, and the greater will be the quantity demanded. POSITIVELY RELATED
- CETERIS PARIBUS: All other things being constant

Demand For A Bond Bd

Price of the Bond	Quantity Demanded	Interest Rate
$950	$100 billion	5.3%
$900	$200 billion	11.1%
$850	$300 billion	17.6%
$800	$400 billion	25%
$750	$500 billion	33.3%

- Consider a deep discount bond of Face Value $1000. Maturity of 1 year.
- At a price of $950, let us say the quantity demanded, Bd, is $100 b.
- At a lower price of $900, the Bd would be $200 b, ceteris paribus. And so on.
- Alternatively, as interest rate rises, Bd increases

Supply of Bond Bs

Price of the Bond	Quantity Supplied	Interest Rate
$750	$100 billion	33.3%
$800	$200 billion	25%
$850	$300 billion	17.6%
$900	$400 billion	11.1%
$950	$500 billion	5.3%

- At a price of $750, interest of 33.3%, let us say the bond issuers are fine with raising Bonds/supply Bonds of $100 b.
- At a lower interest rate or higher price, more companies will want to raise money, the quantity supplied Bs would be more, ceteris paribus. And so on.

Supply of Bonds

3 main factors determine the supply for a particular asset

1. Expected profitability of investment opportunities: Money raised through bonds is invested in business. If the expected profitability from the project is higher (economy is doing well), bond seller would want to raise more bonds at the same price. So, bond supply will increase. Ceteris Paribus. Bs shifts to right. Opposite will happen in a recession.
2. Expected inflation: Means buyers will demand a higher interest rate in future. Makes sense to raise funds now and lock-in cheaper rates. Bond supply increases. Bs shifts to right.
3. Government Budget

- There could be other factors also – we discuss 3 main ones.

Supply of Bonds

3 main factors determine the supply for a particular asset

3. Government Budget:

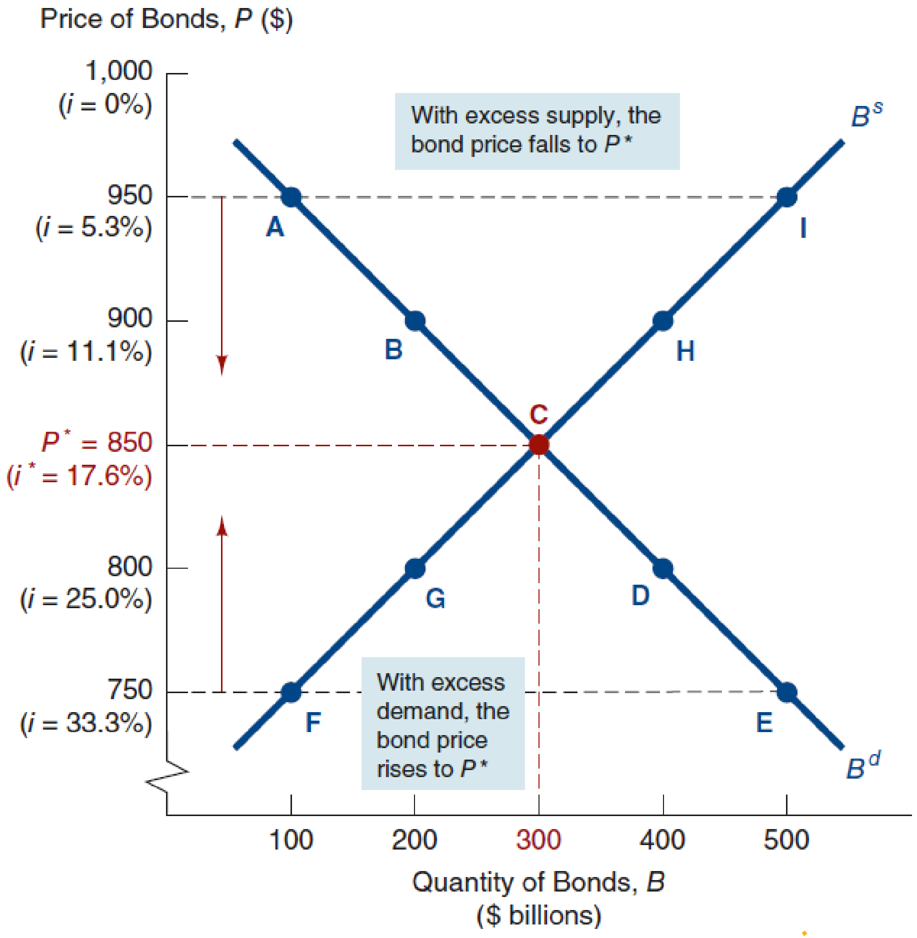
Budget Deficit = Govt Revenue – Govt Expenditure.

Deficit filled up by selling Govt bonds and raising money. So higher the deficit, higher is the amount of govt bonds sold at the same price. Bs moves to right.

- ALL THIS ANALYSIS ASSUMES CETERIS PARIBUS – ALL OTHER THINGS REMAINING CONSTANT.

Bs, Bd Equilibrium

- Price/ interest rate taken as the independent variable. Plotted on Y-axis. Due to old convention in Eco.
- Logic goes as: When the P or int rate increases/decreases then Bs or Bd increases or decreases (as the case may be). ceteris paribus.
- Equilibrium when Bd=Bs. At this price the market clears.
- Excess demand & supply zones.



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Movement Along the Curve

Shift of the Demand / Bd Curve

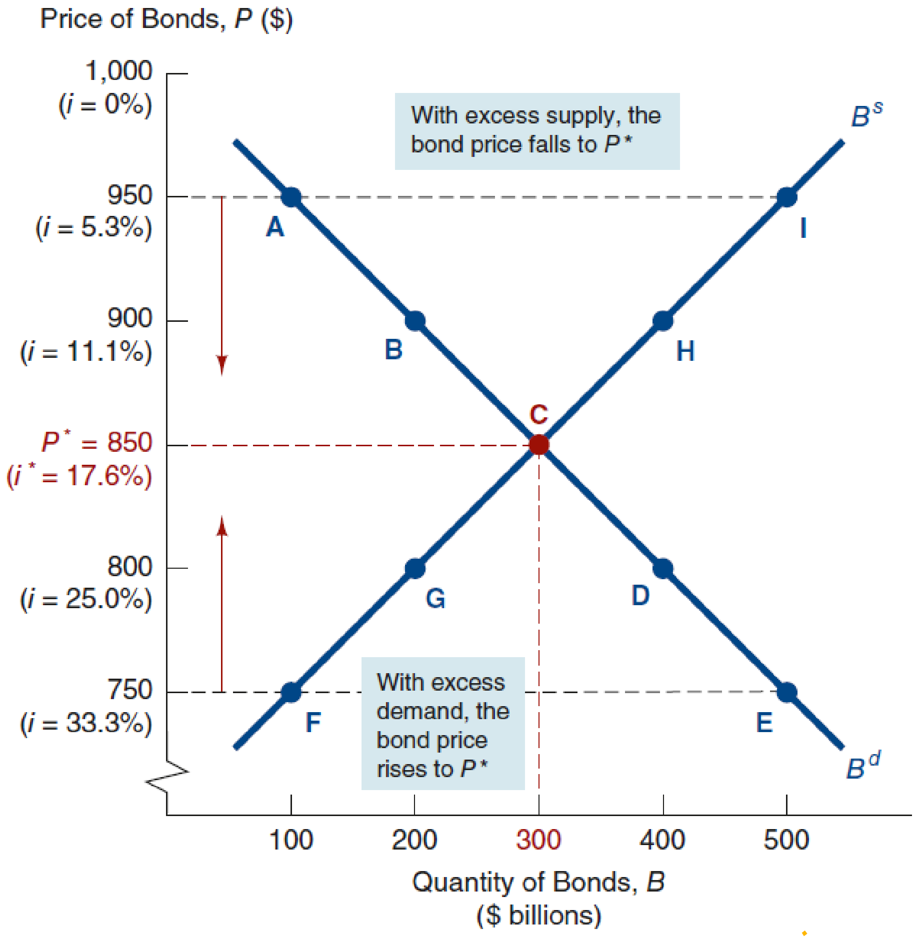
Bs, Bd Equations

- Bd = f(Price/ interest rate, Wealth, Relative [Expected return, Risk, Liquidity]). OR
- Bd = f(Price/ interest rate) ALL ELSE (i.e., Wealth, Relative [Expected return, Risk, Liquidity]) remaining constant. ceteris paribus.

- Bs= f (Price/interest rate, Expected profitability of investment opportunities, expected inflation, expected government deficit) OR
- Bs = f(Price/ interest rate) ALL ELSE (i.e., Expected profitability, expected inflation, expected govt deficit) remaining constant. ceteris paribus.

Movement Along the Curve

- Bs or Bd = f (Price or Interest Rate). Holding all else constant.
- If interest rates go up from 17.6% to 25%, Bd will go up from 300 to 400. Bs will go down from 300 to 200.
- Deficit & excess zones



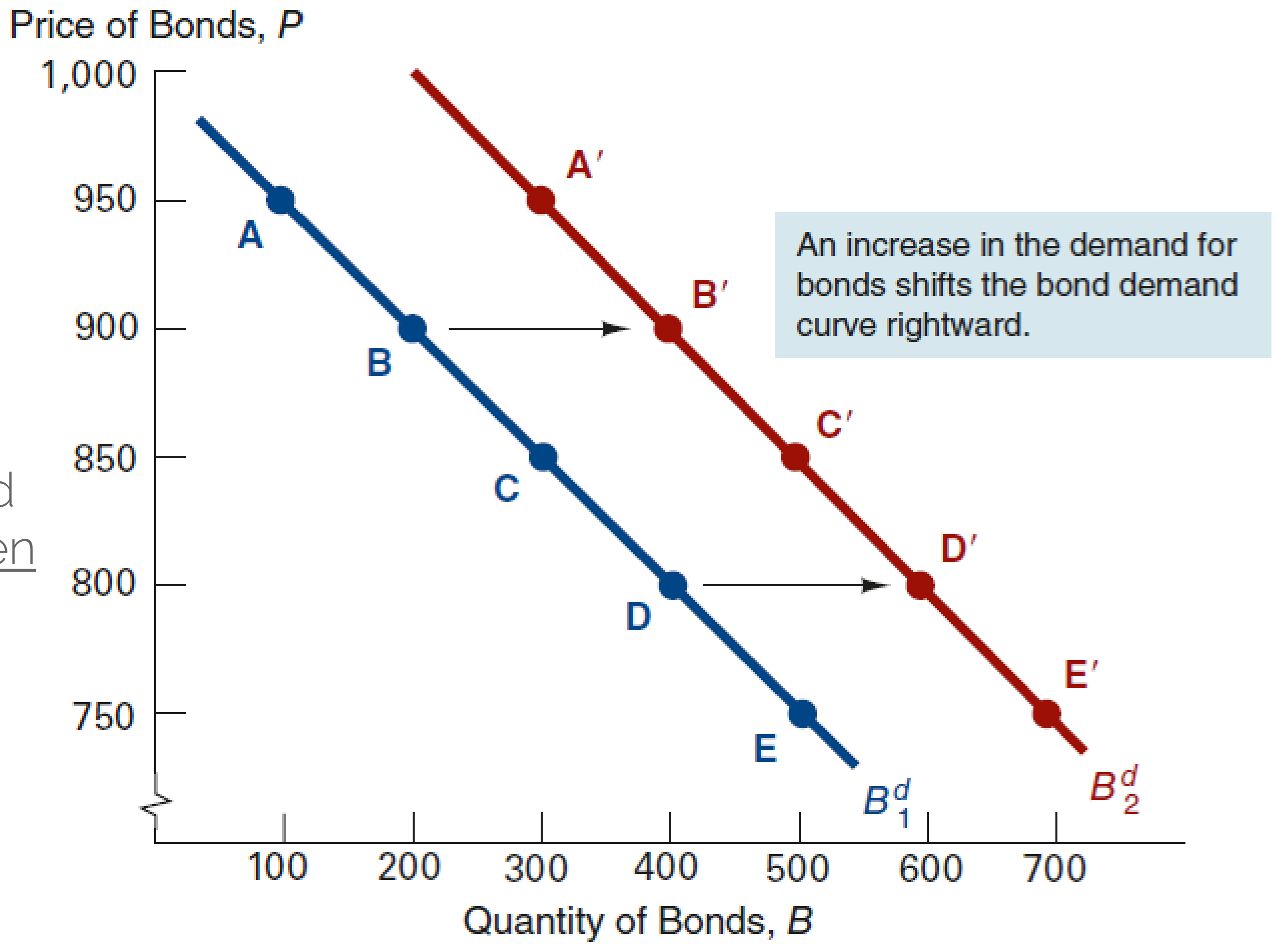
Shift of the Curve

- Change in Bs or Bd when one of the other <u>variables other than Price/ Interest rate changes.</u>
- Wealth or Expected return/Risk/Liquidity Relative to other assets change - then <u>for the same Price/Int rate</u>, the demand for bonds/ **Bd changes** and hence the curve shifts.

- Expected profitability, expected inflation. Govt deficit change - then <u>for the same Price/Int rate</u>, the supply of bonds / **Bs changes** and hence the curve shifts

Shift of Bd Curve: Wealth Changes

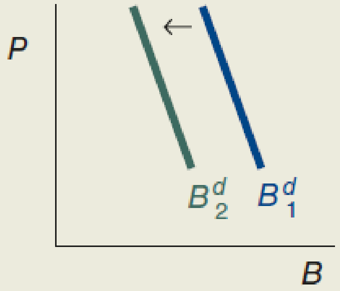
- More bonds demanded at each price point when wealth increases.
- At 900, demand B' instead of B
- At 800, demand D' instead of D.
- Curve shifts to right



Shift of Bd Curve: Wealth Changes

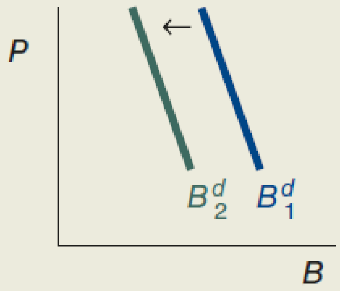
- Economy doing well. Wealth and income increasing. Business cycle expansion. More bonds demanded at each price point. Bd curve shifts to right.
- Economy in recession. Wealth & income decreasing. Less bonds demanded at each price point. Bd curve shifts to left.
- If propensity to save increases, people save more. In times of uncertainty. Feeling pessimistic. Then wealth increases, more demand for bonds at each price point, Bd shifts to right.

Shift of Bd Curve: Expected Interest Rate Changes

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Expected interest rate	↑	↓	

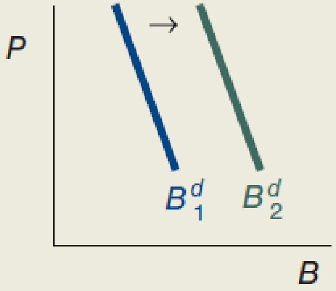
- Higher expected interest rates in the future lower the expected <u>return</u> for long-term bonds, decrease the demand for bonds at each price point, and shift the demand curve to the left. Lower expected interest rates in the future shift Bd to right
- Higher expected returns from other alternative assets (like stocks) vs. bonds, Bd goes down at each price point. Curve shifts to left.

Shift of Bd Curve: Relative Risk Changes

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Riskiness of bonds relative to other assets	↑	↓	

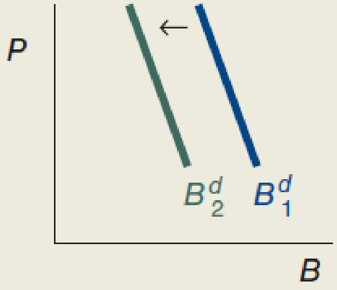
- Higher riskiness (v0latility in prices) of bonds relative to other assets lowers the demand for bonds, and shift the demand curve to the left.
- Higher riskiness in alternative assets like stocks makes bonds more attractive, increase the demand for bonds and shift Bd to right

Shift of Bd Curve: Relative Liquidity Changes

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Liquidity of bonds relative to other assets	↑	↑	

- Increased liquidity of bonds make it more preferred compared to alternative investments. Demand increases. Shift Bd to right.
- Increases in liquidity of alternative assets lower bond demand. Shift Bd to left

Shift of Bd Curve: Expected increase in Inflation

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Expected inflation	↑	↓	

- Inflation means value of other assets (like real estate) will increase. Relatively more attractive in nominal value.
- Inflation will also eat into the value of bond coupons and maturity value. Hence relative attractiveness of bonds decreases.
- Bd will shift to left

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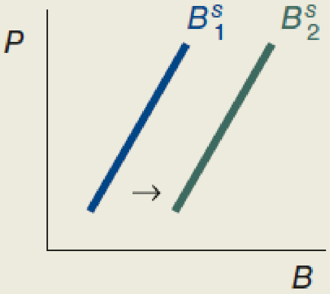
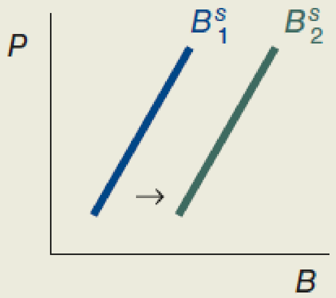
Shift of the Supply / Bs Curve

Shift of Bs Curve: Expected increase in Inflation

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Expected inflation	↑	↑	

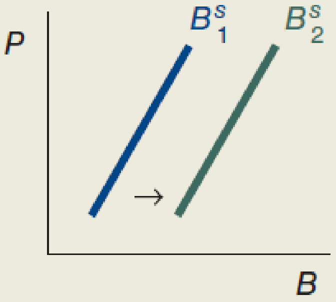
- Expected increase The real cost of borrowing is more accurately measured by the real interest rate, which equals the (nominal) interest rate minus the expected inflation rate. For a given interest rate (and bond price), when expected inflation increases, the real cost of borrowing falls; hence the quantity of bonds supplied increases at any given bond price. Bs will shift to right

Shift of Bs Curve: Expected increase in Inflation

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Expected inflation	↑	↑	

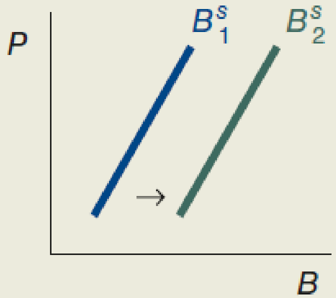
- <u>Alternatively</u>, it inflation is expected to increase, bond buyers will demand a higher coupon rate in the future. So the bond issuer would like to sell more today at a lower coupon i.e. lock in the lower coupon.- Hence the quantity of bonds supplied increases at any given bond price. Bs will shift to right

Shift of Bs Curve: Expected increase in Profitability

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Profitability of investments	↑	↑	

- The more profitable plant and equipment investments that a firm expects it can make, the more willing it will be to borrow to finance these investments.
- Hence Bs will shift to right when economic expansion is happening.
- In a recession, when far fewer profitable investment opportunities are expected, the supply of bonds falls, and the supply curve shifts to the left.

Shift of Bs Curve: Expected increase in Deficit

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Government deficit	↑	↑	

- If government deficit increases, the gap between Govt revenue and Govt expenses increases.
- The government will need more money to fill the gap. It will borrow this money by issuing more bonds in the market. Bs will shift to right

Shift of the Supply / Bs Curve AND Demand/ Bd curve together

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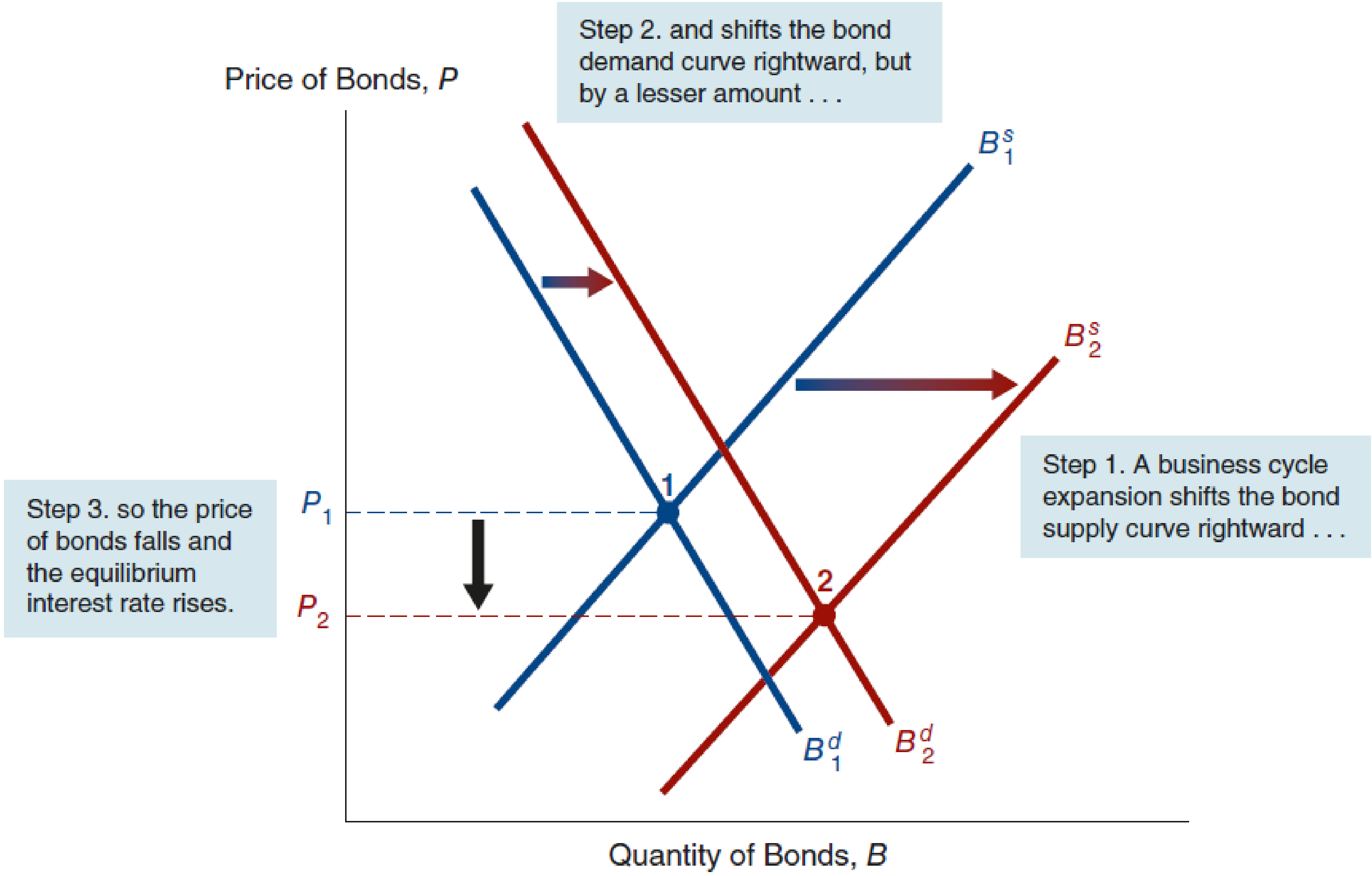
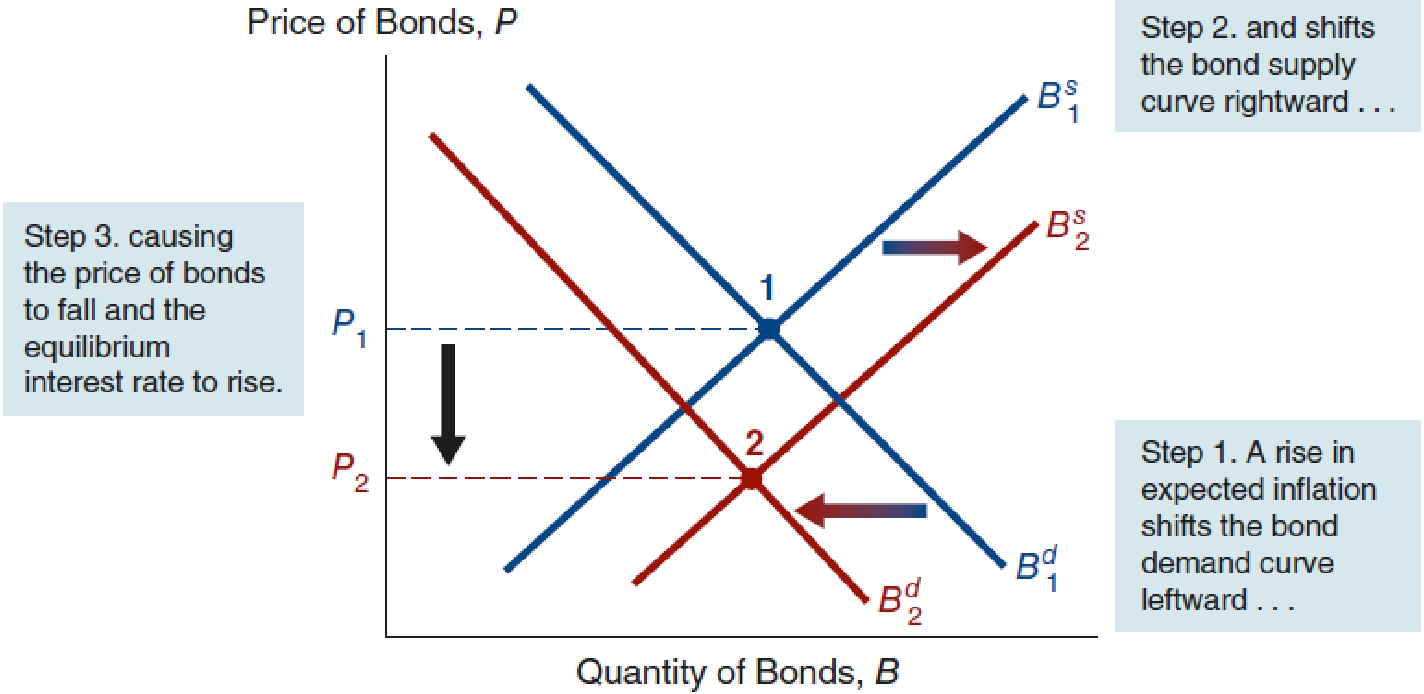
Response to Business Cycle Expansion



Response to Change in Expected Inflation



Revisiting Imp Terms

- Interest Rate: How many?
  - Simple, Compounding, Reducing Balance
  - Short-term, long-term interest rates in the economy
  - Fixed rate, Floating rate
  - Layman Language: What is earned or paid above principal/loan
- YTM Yield to Maturity:
  - Layman Language: Rate of Earning from varying cash flows.
  - Rate at which each 'smaller packets' of Principal amount are compounded (refer earlier discussion related to bonds).
  - Discount rate at which NPV of CFs = Market Price.

Slide included at the beginning of PPT 6 also

Discount Rate/ Bond Valuation

- Example: Face value = ₹100, Coupon = 8%, Maturity = 5 years, Market price = ₹102. Will you buy it?
- A fair required return expected is 7%. Opportunity Cost. Discount Rate?
- Valuation:

$$P = \sum_{t=1}^{5} \frac{8}{(1.07)^t} + \frac{100}{(1.07)^5}$$

- P (**Bond Value** at a discount rate = Fair required return) = INR 104
- Market P of 102 is < **Bond Value** of 104. Hence Undervalued. May be a good buy.

Slide included at the beginning of PPT 6 also

Discount Rate/ Bond Valuation

Comparing 2 Bonds

- Should you buy a XYZ Finance NCD at a price of INR X or buy a GoI 10-year bond at a price of Y? Face Value & Coupon Rate is given.
- Will you calculate & compare YTM & Choose the bond with higher YTM.
  - Yes/No. Why?

Bond	Yield
AAA company	7.5%
Distressed company	14%

- How do you take the risk (default + liquidity) into account?
- Choose a discount rate = risk free rate + risk premium/spread (based on the risk rating of the bond).

Slide included at the beginning of PPT 6 also

Discount Rate/ Bond Valuation

- Comparing two bonds
  - Calculate 'Bond Value' based on a '**chosen discount rate**'.
  - If 'Bond Value' is > Market Price of bond, it is Undervalued
  - If 'Bond Value' is < Market Price of bond, it is Overvalued
- Any Other Factors to be considered??
- Yes. Sensitivity to change in Interest Rates. Depends on ?
- Compare Bonds with similar
  - Maturity
  - Credit Rating
  - Liquidity
  - Other characteristics (Senior/Junior, Secured/Unsecured etc.)

Slide included at the beginning of PPT 6 also

Discount Rate/ Bond Valuation

Example: Market Yield for Different Rated Bonds

Bond Rating	Market Yield
Government	6.8%
AAA	7.6%
AA	8.3%
A	9.2%

The discount rates will also follow a trend similar to Market Yield- Low for Govt and higher as we move to riskier bonds with lower credit ratings

Slide included at the beginning of PPT 6 also

Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	
C	D	D	In default
/			
/			

"Junk"

Thank You!